

A Resolution of Erdos 374

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Abstract

For a composite integer m , let $F(m)$ be the least number of distinct positive-index factorials, the largest being $m!$, whose product is a square, and put $D_6 = \{m : F(m) = 6\}$. We prove that D_6 has positive lower density. More precisely, we construct a squarefree set $\mathcal{A}$ of positive natural density $\delta_{\mathcal{A}}$ and show that almost every member of $\mathcal{A}$ having a prime factor greater than $m^{99/100}$ belongs to D_6 . This gives lower density at least $\delta_{\mathcal{A}} \log(100/99)$. The argument combines a uniform lower bound for the logarithmic mass of valuation-one prime divisors of a long interval with a large-sieve count anchored at the largest prime factor of the endpoint. The analytic inputs are the prime number theorem, a published reciprocal-prime equidistribution estimate, Harman's almost-all short-interval theorem, Weil's bound, and the analytic large sieve. We also determine the order of growth of the classes $D_k = \{m : F(m) = k\}$ for $3 \leq k \leq 6$: $D_3(X) \asymp \sqrt{X}$, while D_4 , D_5 and D_6 have positive lower density. All these results are verified in Lean.

1 Introduction and statement

For $m > 1$, define $F(m)$ to be the least $k \geq 2$ for which there are positive integers

$$1 \leq a_1 < a_2 < \cdots < a_k = m, \quad a_1! a_2! \cdots a_k! \text{ a square.}$$

Set $F(m) = \infty$ if no such representation exists, and let

$$D_6 = \{m > 1 : F(m) = 6\}.$$

Erdős and Graham studied the corresponding nested sets of endpoints representable with at most k factorials. They showed, in particular, that every composite endpoint admits at most six factors, and asked whether D_6 has positive lower density [1, pp. 341, 354]. The minimum convention used here agrees with their difference-set convention for $m > 1$. Including $1!$ does not change a minimum: it may be removed, and a single nontrivial factorial is not a square.

For $n \geq 1$, let

$$\text{sf}(n) = \prod_{v_p(n) \text{ odd}} p, \quad q_a = \text{sf}(a!), \quad a \geq 0.$$

Thus $q_0 = q_1 = 1$. Define

$$\mathcal{A} = \left\{ m \geq 1 : m \text{ is squarefree and } \gcd(m, q_a)^2 \leq q_a \text{ for every } a \geq 2 \right\}. \quad (1.1)$$

For $0 < \alpha < 1/2$, write

$$\mathcal{B}_{\alpha} = \{m \in \mathcal{A} : P^+(m) > m^{1-\alpha}\}, \quad (1.2)$$

where $P^+(m)$ is the largest prime factor, and $P^+(1) = 1$.

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Theorem 1.1 (Main statement). *The set $\mathcal{A}$ has a positive natural density δ_A . Moreover,*

$$\#(D_6 \cap \mathcal{B}_{1/100} \cap [1, X]) = (\delta_A \log(100/99) + o(1))X. \quad (1.3)$$

In particular,

$$\liminf_{X \rightarrow \infty} \frac{\#(D_6 \cap [1, X])}{X} \geq \delta_A \log(100/99) > 0.$$

The conclusion does not assert existence of the natural density of the whole set D_6 , nor determine the other endpoint-counting functions in Problem 374. No effective value of δ_A , and no effective threshold for the assembled argument, is claimed.

1.1 Structure and provenance of the argument

Write

$$P_h(t) = \prod_{i=0}^{h-1} (t - i).$$

For a five-factor representation $a!b!c!d!m!$ with $a < b < c < d < m$, put

$$h = c - b, \quad \ell = m - d.$$

Cancellation of square factors gives

$$a!b!c!d!m! \text{ square} \iff q_a P_h(c) P_\ell(m) \text{ square}. \quad (1.4)$$

For four factors $b!c!d!m!$, use $q = 1$ and formally $a = 0$. The two-interval formulation, the paired-avoidance strategy, and the six-factor construction are classical [1]. Erdős–Graham also established an almost-all result on the fixed-cofactor family $13p$; a fixed-cofactor prime family has density zero among all integers.

The two main estimates here have different roles. A valuation-one prime-mass lemma shows that a polynomially long lower interval cannot be matched by a much shorter upper interval. A prime-anchor sieve then counts all remaining nonpaired representations when the endpoint has a very large prime factor. The class $\mathcal{A}$ excludes the paired representations exactly. The density calculation for $\mathcal{B}_\alpha$ is by cofactor summation, not an independence assertion.

The reciprocal-prime method is taken from the published theorem of Matomäki, Radziwiłł, Shao, Tao, and Teräväinen [2]. Its use in interval anatomy in [7] provided methodological guidance. The proof below is self-contained apart from the external inputs stated in Section 2; none of the exploratory finite tables or earlier fixed-gap integral-point results is a logical dependency. Priority for the particular combination of estimates has not been established by a comprehensive literature review.

2 Notation and external inputs

All logarithms are natural. Intervals counted over integers are understood to include only their integer points. Write $\vartheta(x) = \sum_{p \leq x} \log p$, and let $\pi(x)$ count primes. All assertions with “sufficiently large” allow a threshold depending on explicitly fixed exponents. We use the quantitative prime number theorem

$$\vartheta(x) = x + O(x \exp(-c\sqrt{\log x})) \quad (2.1)$$

for an absolute $c > 0$, as well as $\pi(x) \sim x / \log x$; see, for example, [4].

2.1 Reciprocal-prime equidistribution

We use the following published result, with $j = 2$ throughout.

Proposition 2.1 ([2, Proposition 1.13(ii)]). *Fix $\varepsilon > 0$, $A_0 > 0$, and a natural number j . Let $P \geq 2$ and $I \subseteq [P, 2P]$ be an interval. For real parameters*

$$M, N = O\left(\exp((\log P)^{3/2-\varepsilon})\right)$$

and a smooth $\mathbb{Z}^2$ -periodic function W , one has

$$\sum_{p \in I} W\left(\frac{N}{p}, \frac{M}{p^j}\right) = \int_I W\left(\frac{N}{t}, \frac{M}{t^j}\right) \frac{dt}{\log t} + O_{\varepsilon, A_0, j}\left(\|W\|_{C^3} \frac{P}{(\log P)^{A_0}}\right). \quad (2.2)$$

Here the norm is a sum of supremum norms of derivatives of orders at most three.

The extra displayed dependence on the fixed j is harmless; only $j = 2$ is used. The numbering above is that of the published paper, not the initial preprint. Crucially, the main term is an integral *along the same reciprocal curve*. It does not state independence of its coordinates. The explicit norm factor allows the cutoff to depend on X .

2.2 Almost-all prime intervals

Harman's theorem [3] implies that, for every fixed $\theta > 1/10$, all but $o(X)$ integers $m \leq X$ have a prime in

$$(m - m^\theta, m]. \quad (2.3)$$

The exceptional estimate from the cited proof is $O_\theta(X/\log X)$; only $o(X)$ is needed for Theorem 1.1. To obtain the integer backward form from a real forward formulation, choose $\theta_0 \in (1/10, \theta)$. On $X < m \leq 2X$, set $L = (2X)^{\theta_0}$ and $C = \lceil L \rceil + 2$. A bad integer m makes every start in $[m - C, m - C + 1/2]$ bad for a forward interval of length x^{θ_0} . For large X these intervals lie within (2.3), and the half-unit sets of starts are disjoint. Harman's exceptional-measure estimate therefore bounds the bad integers; dyadic summation gives the stated form. See also the discussion of the exceptional-set strength in [7, Remark 2.4].

2.3 Quadratic characters and the large sieve

Let χ_p be the quadratic character modulo an odd prime, with $\chi_p(0) = 0$. If $f \in \mathbb{F}_p[z]$ is not a scalar times a square, Weil's bound, with the zero corrections from any removed square factors, gives

$$\left| \sum_{z \bmod p} \chi_p(f(z)) \right| \ll (\deg f) \sqrt{p}. \quad (2.4)$$

We use the usual coefficient-uniform form [5].

For a Δ -separated set $\mathcal{T} \subset \mathbb{R}/\mathbb{Z}$, the analytic large sieve gives

$$\sum_{\theta \in \mathcal{T}} \left| \sum_{1 \leq n \leq L} a_n e^{2\pi i n \theta} \right|^2 \ll (L + \Delta^{-1}) \sum_{1 \leq n \leq L} |a_n|^2. \quad (2.5)$$

We only need $\Delta \geq 1/(4L)$ [4, 6]. All applications below check good reduction and keep zero among the allowed residues for integer squares.

3 A positive-density set excluding paired representations

Lemma 3.1 (Size of factorial square classes). *As $a \rightarrow \infty$,*

$$\log q_a \geq (1/2 + o(1))a, \quad \omega(q_a) = O(a/\log a).$$

In particular $q_a \geq \exp(a/3)$ for all sufficiently large a .

Proof. Every prime $a/2 < p \leq a$ occurs once in $a!$, so its product divides q_a . Apply (2.1). Every prime factor of q_a is at most a , giving the second assertion by the prime number theorem. $\square$

Lemma 3.2. *The set $\mathcal{A}$ in (1.1) has a positive natural density δ_A and is closed under taking squarefree divisors.*

Proof. Failure of the a -th gcd condition forces divisibility of m by some $d \mid q_a$ with $d > \sqrt{q_a}$. Uniformly for every $X \geq 1$, the proportion of integers $m \leq X$ having this failure is at most

$$\sum_{\substack{d \mid q_a \\ d > \sqrt{q_a}}} \frac{1}{d} \leq \frac{2^{\omega(q_a)}}{\sqrt{q_a}} \leq \exp(-(1/4 - o(1))a). \quad (3.1)$$

Here $\lfloor X/d \rfloor \leq X/d$ is why the estimate is uniform in the prefix. The bound is summable in a .

For fixed B , squarefree integers with no prime factor at most B have density

$$\delta_B = \frac{6}{\pi^2} \prod_{p \leq B} (1 + 1/p)^{-1} \asymp \frac{1}{\log B}.$$

They satisfy every gcd condition with $a \leq B$. Removing all violations with $a > B$ costs $O(\exp(-cB))$, by (3.1), which is smaller than δ_B for large B . Thus $\mathcal{A}$ has positive lower density.

For each fixed number of gcd conditions, the intersection with the squarefree integers has a natural density. One may see this by imposing finitely many prime-square exclusions and then using the tail bound $\sum_{p > T} p^{-2}$; the other conditions are periodic. Equation (3.1) gives uniform convergence of these finite-condition densities to the density of $\mathcal{A}$. Finally, if $d \mid m$ is squarefree, then $\gcd(d, q_a) \leq \gcd(m, q_a)$ for every a , proving closure under divisors. $\square$

Lemma 3.3 (Paired exclusion). *If $m \in \mathcal{A}$ and $1 \leq c < m$, then $q_a c m$ is not a square for any $a \geq 1$ compatible with the factorial ordering. The same holds for cm .*

Proof. Let $g = \gcd(m, q_a)$. Since m, q_a are squarefree, squarefreeness would force

$$c = \frac{q_a m}{g^2} z^2$$

for some integer $z \geq 1$. The definition of $\mathcal{A}$ gives $g^2 \leq q_a$, hence $c \geq m$, a contradiction. For $a = 1$ or multiplier 1, the same reasoning gives $c = m z^2$. $\square$

This excludes precisely the gap pair $(h, \ell) = (1, 1)$ in the four- and five-factor problems. It does not exclude arbitrary shorter representations.

Lemma 3.4 (Cofactor inheritance). *Fix $0 < \alpha < 1/2$. For all sufficiently large $m = rP$ with $P > m^{1-\alpha}$ prime,*

$$m \in \mathcal{A} \iff r \in \mathcal{A}.$$

Proof. We have $P > r$ and $P^2 > m$. Thus squarefreeness of rP is equivalent to that of r . If $a < P$, then $P \nmid q_a$, so $\gcd(rP, q_a) = \gcd(r, q_a)$. If $a \geq P$, Lemma 3.1 gives

$$q_a \geq \exp(a/3) \geq \exp(P/3) > m^2$$

for sufficiently large m . Both gcd conditions are then automatic. This proves the equivalence for *all* factorial indices, not merely those below m . $\square$

Proposition 3.5 (Density of the large-prime subfamily). *For $0 < \alpha < 1/2$,*

$$\#(\mathcal{B}_\alpha \cap [1, X]) = \left(\delta_A \log \frac{1}{1-\alpha} + o(1) \right) X. \quad (3.2)$$

Proof. Put $\beta = (1 - \alpha)/\alpha > 1$. The inequality $P > (rP)^{1-\alpha}$ is equivalent to $P > r^\beta$. The large prime is unique. Lemma 3.4 therefore gives, up to finitely many initial exceptions,

$$\#(\mathcal{B}_\alpha \cap [1, X]) = \sum_{\substack{r \leq X^\alpha \\ r \in \mathcal{A}}} (\pi(X/r) - \pi(r^\beta)) + O(1). \quad (3.3)$$

Boundary values are interpreted using $\pi(t) = \#\{p \leq t\}$, so the strict inequality at the lower endpoint is correct.

The second sum is $O_\alpha(X/\log X)$. Indeed, the range $r \leq X^{\alpha/2}$ is $O(X^{1/2})$ by the trivial bound $\pi(r^\beta) \leq r^\beta$, and on the other range the elementary prime upper bound saves a factor $\log X$.

In the first sum, $X/r \geq X^{1-\alpha}$, so the prime number theorem has a uniform relative $o(1)$ error. Partial summation using $\#(\mathcal{A} \cap [1, t]) = \delta_A t + o(t)$ gives

$$\begin{aligned} \sum_{\substack{r \leq X^\alpha \\ r \in \mathcal{A}}} \frac{1}{r \log(X/r)} &= \delta_A \int_1^{X^\alpha} \frac{dt}{t \log(X/t)} + o(1) \\ &= \delta_A \log \frac{1}{1-\alpha} + o(1). \end{aligned}$$

For error control, fix T_0 so the counting-function error is at most ϵt above T_0 . The integral of t times the absolute derivative of $1/(t \log(X/t))$ is $O_\alpha(1)$; the finite part tends to zero, and then ϵ tends to zero. Substituting in (3.3) proves the result. Prime endpoints correspond to $r = 1$ and may be removed without changing the density. $\square$

4 Valuation-one prime mass in a long interval

The following lemma supplies the uniform comparison between the two interval lengths.

Lemma 4.1 (Valuation-one prime mass). *Fix $0 < \gamma < 1$. For all sufficiently large X depending on γ , suppose integers $0 \leq a < b < c \leq X$ satisfy*

$$h = c - b \geq X^\gamma, \quad a \leq 10h \log X.$$

There is a squarefree integer R , composed of primes $p > a$ with $v_p(P_h(c)) = 1$, such that

$$\log R \geq h/4096. \quad (4.1)$$

The assertion is uniform in a, b, c in the displayed range.

4.1 The dense range

Put $U = \log X$ and $P = hU^2$. Suppose $b \leq 100P$. Let

$$J = (\max(b, c/2), c], \quad L = |J| = \min(h, c/2).$$

Then $L \geq h/2$, while $c \leq 101hU^2$ gives $L \geq c/(202U^2)$. Since $c \geq X^\gamma$, the quantitative prime number theorem gives, uniformly,

$$\vartheta(c) - \vartheta(c - L) = L + o(L) \geq h/4.$$

Both endpoints are on the c scale, and the error divided by L is $O(U^2 \exp(-c_\gamma \sqrt{U})) = o(1)$. Every prime in J exceeds $b > a$ and $c/2$, and so occurs exactly once in $P_h(c)$. This proves more than (4.1). It also covers cases in which the auxiliary quantity P exceeds X .

4.2 Cutoffs in the sparse range

Assume henceforth $b > 100P$. Then, for sufficiently large X ,

$$P > a, h, \quad c < 2b, \quad 2P < X.$$

Set

$$\delta = \frac{h}{4P} = \frac{1}{4U^2}.$$

Choose a smooth 1-periodic $0 \leq \phi_\delta \leq 1$ supported where

$$\tfrac{1}{2}\delta \leq 1 - \{x\} \leq \delta,$$

and equal to one where $1 - \{x\} \in [5\delta/8, 7\delta/8]$. A fixed bump on $[1/2, 1]$, rescaled and periodized, gives

$$\int_0^1 \phi_\delta(x) dx \geq \delta/4, \quad \|\phi_\delta\|_{C^3} = O(\delta^{-3}) = O(U^6). \quad (4.2)$$

Choose a fixed smooth periodic $0 \leq \psi \leq 1$, supported where the fractional part lies in $(1/20, 17/20)$ and equal to one on $[9/100, 81/100]$.

If $P \leq p \leq 2P$ and $\phi_\delta(b/p) > 0$, the next multiple of p after b lies at distance between $\delta p/2$ and δp , hence within $h/2$. It is the unique multiple in $(b, b+h]$ because $p > h$.

If $P > \sqrt{2b}$, then $p^2 > c$ and its valuation is one. In this case take $W(x, y) = \phi_\delta(x)$. Otherwise take

$$W(x, y) = \phi_\delta(x)\psi(y).$$

A positive value of $W(b/p, b/p^2)$ then places the next multiple of p^2 at distance greater than $3p^2/20 > h$ after b . Thus the selected multiple of p again has valuation exactly one. Every selected prime exceeds a . In both cases,

$$\|W\|_{C^3} = O(U^6). \quad (4.3)$$

Remark 4.2. The second coordinate is essential when p^2 can lie below c . A cutoff that only detects a multiple of p does not distinguish valuation one from valuation two. For a finite check consistent with $P = h(\log X)^2$, take $p = 1\,000\,003$, $h = 1000$, $b = p^2 - 250$, and $X = c = b + h$. Rational logarithm enclosures give $P < p < 2P$, $b > 100P$, and

$$\frac{5}{8} < \frac{1 - \{b/p\}}{\delta} = \frac{P}{p} < \frac{7}{8}.$$

Thus $\phi_\delta(b/p) = 1$, while $p^2 \in (b, c]$ and $v_p(P_h(c)) = 2$. Since $\{b/p^2\} > 17/20$, the second cutoff vanishes. This checks strict plateau membership, not just membership in the closed support; it does not verify the eventual starting threshold of Lemma 4.1. No assertion about odd valuation below uses the first coordinate alone when p^2 can lie below c .

4.3 The correlated continuous main term

We prove in either sparse subcase that

$$I = \int_P^{2P} W\left(\frac{b}{t}, \frac{b}{t^2}\right) \frac{dt}{\log t} \geq \frac{h}{320 \log(2P)}. \quad (4.4)$$

Make the substitution $s = b/t$ and write $A = b/P > 100$. Then $s \in [A/2, A]$, and

$$\frac{b}{s^2 \log(b/s)} \geq \frac{b}{A^2 \log(2P)}. \quad (4.5)$$

If no second cutoff is needed, there are at least $A/2 - 2 \geq A/3$ complete unit cells, each contributing at least $\delta/4$ by (4.2). This gives a stronger bound than (4.4).

For the other subcase, put $B = b/P^2 \geq 1/2$. As s varies, $y = s^2/b$ ranges over $[B/4, B]$, of length $3B/4 \geq 3/8$. We use the exact elementary fact

$$\inf_x |[x, x+L] \cap \{y : \{y\} \in [1/10, 4/5]\}| = \frac{7}{10} \lfloor L \rfloor + \max\left(0, \{L\} - \frac{3}{10}\right). \quad (4.6)$$

Indeed every complete period contributes $7/10$, and the remaining interval of length $\{L\}$ can lose at most the complementary arc of length $3/10$. In particular the right side is at least $L/5$ for $L \geq 3/8$.

Thus the good y -measure is at least $3B/20$. Since

$$\frac{ds}{dy} = \frac{\sqrt{b}}{2\sqrt{y}} \geq \frac{\sqrt{b}}{2\sqrt{B}},$$

the corresponding good set in s has measure at least $3A/40$. Discard the two possible partial unit cells at the endpoints. At least

$$3A/40 - 2 \geq A/20$$

complete unit cells meet the good set. On each such cell the variation of s^2/b is at most $2/P < 1/100$ for large X . Since one point has fractional part in $[1/10, 4/5]$, the whole cell lies in the plateau $[9/100, 81/100]$ of ψ . The margins also prevent wraparound at an integer.

Each of these cells contributes at least $\delta/4$ to the s integral. By (4.5),

$$I \geq \frac{b}{A^2 \log(2P)} \frac{A}{20} \frac{\delta}{4} = \frac{P\delta}{80 \log(2P)} = \frac{h}{320 \log(2P)}.$$

This establishes the lower bound along the actual correlated curve.

4.4 Transfer to primes and conclusion

Apply Proposition 2.1 with

$$\varepsilon = 1/4, \quad A_0 = 20, \quad j = 2, \quad M = N = b.$$

Because $P \geq X^\gamma U^2$,

$$\log P \geq \gamma U, \quad b \leq X \leq \exp((\log P)^{5/4})$$

for all sufficiently large X depending only on γ .

$$\frac{(\log P)^{5/4}}{\log X} \geq \gamma^{5/4} (\log X)^{1/4} \longrightarrow \infty \quad (X \rightarrow \infty, \gamma > 0 \text{ fixed}).$$

Thus its phase hypothesis holds uniformly. Equations (4.3) and (2.2) bound the error by

$$O_\gamma\left(U^6 \frac{hU^2}{U^{20}}\right) = O_\gamma(h/U^{12}).$$

In the sparse range, $\log(2P) \leq U$, so the error is $o(h/\log(2P))$. Therefore

$$\sum_{P \leq p \leq 2P} W(b/p, b/p^2) \geq \frac{h}{640 \log(2P)}.$$

The weights are in $[0, 1]$. Take R to be the product of the primes with positive weight. The valuation statement proved above holds for every such prime, and

$$\log R \geq \frac{h \log P}{640 \log(2P)} \geq h/1280.$$

This implies the weaker constant in (4.1) and completes the proof of Lemma 4.1.

Corollary 4.3 (Comparison of interval lengths). *Fix $\gamma > 0$. For sufficiently large m , every four- or five-factor square representation with $h \geq \ell$ and $h \geq m^\gamma$ satisfies*

$$h \leq 4096 \ell \log m. \quad (4.7)$$

Proof. Squareness in (1.4) implies $q_a \mid P_h(c)P_\ell(m)$, hence $q_a \leq m^{h+\ell}$. Lemma 3.1 gives

$$a \leq 3(h + \ell) \log m + O(1) \leq 10h \log m$$

eventually. Apply Lemma 4.1 with $X = m$ (and $a = 0$ for four factors). Each prime in R is absent from $a!$ and occurs once in the lower interval. To make the total product a square it must divide the upper interval product. Thus

$$R \mid P_\ell(m), \quad h/4096 \leq \log R \leq \log P_\ell(m) \leq \ell \log m.$$

No restriction on the separation of the intervals is used. $\square$

5 The uniform large-prime-anchor sieve

Proposition 5.1. *Fix $\alpha, \eta > 0$ with*

$$3\alpha + 8\eta < 1. \quad (5.1)$$

Among endpoints $m \leq X$ with $P^+(m) > m^{1-\alpha}$, the number admitting a nonpaired five-factor square representation with $h, \ell \leq m^\eta$ is

$$O_{\alpha, \eta} \left(X^{1/2+3\alpha/2+4\eta} (\log X)^2 \right). \quad (5.2)$$

For four factors it is

$$O_{\alpha, \eta} \left(X^{1/2+3\alpha/2+3\eta} \log X \right). \quad (5.3)$$

All bottom indices and all interval separations are included. Squarefreeness of m is not assumed.

Proof. Work first on $X/2 < m \leq X$. Put $H = X^\eta$, $R = X^\alpha$, and $m = rP$ with $P = P^+(m)$. Then

$$r \leq R, \quad P \geq X/(2R), \quad P^2 > m,$$

since (5.1) implies $\alpha < 1/3$. The factorial-size bound used above gives

$$a \leq 6H \log X + O(1). \quad (5.4)$$

For large X , $a, h, \ell < P$. The prime P occurs once in the upper interval, at m , and is absent from $a!$. It must therefore divide a lower-interval member. As $h < P$, this member is unique, so

$$c = jP + t, \quad 1 \leq j < r, \quad 0 \leq t < h. \quad (5.5)$$

If $r = 1$, no such j exists.

After cancelling the two factors P , the square condition is that $f(P)$ is an integer square, where

$$f(z) = q_a r j \prod_{i=1}^{\ell-1} (rz - i) \prod_{\substack{0 \leq v < h \\ v \neq t}} (jz + t - v). \quad (5.6)$$

The two rational root lists, with multiplicity, are

$$\left\{ \frac{i}{r} : 1 \leq i < \ell \right\}, \quad \left\{ \frac{v-t}{j} : 0 \leq v < h, v \neq t \right\}. \quad (5.7)$$

Outside $(h, \ell) = (1, 1)$, f is not a scalar times a square. If $t > 0$, the second list has an unmatched negative root. If $t = 0$ and exactly one list is empty, every root in the nonempty list has odd multiplicity,

so again f is not a scalar times a square. If $t = 0$ and both lists are nonempty, all roots are positive. For every root to occur evenly, the lists would have to coincide, forcing $1/r = 1/j$, impossible because $j < r$. Both lists are empty precisely in the excluded paired case.

Fix a, r, j, t, h, ℓ , and set $L = X/r$. Use auxiliary primes

$$\sqrt{L} < \lambda \leq 2\sqrt{L}.$$

The parameter condition gives

$$\alpha + \eta < (1 - \alpha)/2, \quad \eta < (1 - \alpha)/4. \quad (5.8)$$

Together with (5.4), these imply $H = o(\sqrt{\lambda})$ uniformly. In particular, for all sufficiently large X , uniformly over every tuple,

$$\lambda > \max(a, 2RH), \quad \lambda \geq 64H^2.$$

These are consequences of the fixed parameter inequalities, not additional hypotheses. Thus $\lambda \nmid q_a r j$. Nonzero cross-multiplied differences between roots in (5.7) have absolute value at most $2RH$, so distinct rational roots remain distinct modulo λ ; exact doubled roots remain doubled. The nonsquare property survives reduction.

Apply the squarefree form of Weil's bound to the odd-multiplicity factors, and add at most one unit of error for each removed double root. Since the total degree is at most $2H$, this gives the explicit form of (2.4)

$$\left| \sum_{z \bmod \lambda} \chi_\lambda(f(z)) \right| \leq 2H\sqrt{\lambda}.$$

If z_λ is the number of zero residues, then $z_\lambda \leq 2H$. Hence the number ρ_λ of square-or-zero values satisfies

$$\rho_\lambda = \frac{\lambda + z_\lambda + \sum_z \chi_\lambda(f(z))}{2} \leq \frac{\lambda}{2} + H + H\sqrt{\lambda} \leq \frac{3\lambda}{4},$$

where the last inequality follows from $\lambda \geq 64H^2$ and $H \geq 1$. All estimates are uniform in the tuple, and zero values have been included.

Let $T \subseteq [1, L] \cap \mathbb{Z}$ be the integer arguments where f is a square, and put $J = \#T$. This contains every prime argument arising from the original representation. If $n_\lambda(u)$ is the number of members of T in residue class u , orthogonality and Cauchy-Schwarz give

$$\begin{aligned} \sum_{b=1}^{\lambda-1} \left| \sum_{z \in T} e^{2\pi i b z / \lambda} \right|^2 &= \lambda \sum_u n_\lambda(u)^2 - J^2 \\ &\geq (\lambda/\rho_\lambda - 1)J^2 \geq J^2/3. \end{aligned}$$

The fractions b/λ are separated modulo one by at least $1/(4L)$. By (2.5), their combined energy is $O(LJ)$. The prime number theorem supplies $\gg \sqrt{L}/\log L$ auxiliary primes. Hence, also when $J = 0$,

$$J \ll \sqrt{L} \log L \ll \sqrt{X/r} \log X. \quad (5.9)$$

For each r , there are at most r choices for j , H for ℓ , $O(H^2)$ for (h, t) , and $O(H \log X)$ for a . Summing (5.9) gives

$$O\left(H^4 \sqrt{X} (\log X)^2 \sum_{r \leq R} \sqrt{r}\right) = O\left(\sqrt{X} H^4 R^{3/2} (\log X)^2\right).$$

This proves the dyadic five-factor bound. Omitting the a -sum proves the four-factor bound. Dyadic summation preserves the displayed powers, proving the proposition. $\square$

Remark 5.2 (Uniformity and scope). The polynomial in (5.6) may take square values; the proof only asserts that it is not itself a scalar multiple of a polynomial square. The sieve includes zeros, and its count explicitly pays for all six tuple variables a, r, j, t, h, ℓ . No bound on the numerical height of q_a is needed for good reduction: the auxiliary primes exceed a . This is an application of standard character-sum and large-sieve methods, not a claim that those methods are new.

6 Short upper gaps and completion of the argument

Lemma 6.1 (The second-largest factorial index). *Outside $O(X/\log X)$ integers $m \leq X$ and finitely many initial values, every square product of at least two distinct factorials with largest index m and second-largest index d satisfies*

$$\ell = m - d \leq m^{11/100}.$$

Proof. Let $p_-(m)$ be the largest prime at most m . Bertrand's postulate gives $p_-(m) > m/2$ for large m . If $d < p_-(m)$, that prime occurs once in $m!$ and in none of the other factorials, a contradiction. Thus

$$\ell \leq m - p_-(m).$$

Apply (2.3) with $\theta = 11/100$. The exceptional set depends only on the endpoint m , so the assertion holds for every representation there simultaneously. $\square$

Corollary 6.2. *Outside the same exceptional endpoint set and finitely many initial values, every four- or five-factor representation has*

$$h < m^{3/25}, \quad \ell \leq m^{11/100} < m^{3/25}.$$

Proof. If $h \geq m^{3/25}$, then $h \geq \ell$ for sufficiently large m . Corollary 4.3, with the fixed exponent $\gamma = 3/25$, gives

$$h \leq 4096\ell \log m \leq 4096m^{11/100} \log m < m^{3/25},$$

a contradiction. The repair lemma is uniform, so no union over representations or gaps is needed. $\square$

6.1 Proof of Theorem 1.1

Choose

$$\alpha = 1/100, \quad \theta = 11/100, \quad \eta = 3/25.$$

The parameter comparisons, including the narrowest final saving, are

Comparison	Value or positive margin
$\theta - 1/10$	$1/100$
$\eta - \theta$	$1/100$
$1 - (3\alpha + 8\eta)$	$1/100$
$1 - (1/2 + 3\alpha/2 + 4\eta)$	$1/200$
$1 - (1/2 + 3\alpha/2 + 3\eta)$	$1/8$
$(1 - \alpha)/2 - (\alpha + \eta)$	$73/200$
$(1 - \alpha)/4 - \eta$	$51/400$

Proposition 3.5 gives $\mathcal{B}_{1/100}$ density $\delta_A \log(100/99) > 0$. Delete its prime members, the bad endpoints of Lemma 6.1, and finitely many initial values. By Corollary 6.2, any four- or five-factor representation at a remaining endpoint has both gaps below m^η . The paired case is excluded by Lemma 3.3. Proposition 5.1 bounds all the nonpaired cases by

$$O\left(X^{199/200}(\log X)^2\right) + O\left(X^{7/8} \log X\right) = o(X). \quad (6.1)$$

Two- and three-factor representations can be excluded directly on this population. Let $P = P^+(m) > m^{99/100}$. It occurs once in m , exceeds ℓ , and thus occurs once in $P_\ell(m)$. A two-factor representation would make $P_\ell(m)$ a square, which is impossible. A three-factor representation $a!d!m!$ would make $q_a P_\ell(m)$ square. It forces

$$a \leq 3\ell \log m + O(1) < P.$$

So P is absent from q_a and again has odd valuation in the total product. This also covers $a = 1$.

Finally, for any composite $m = uv$, $u, v \geq 2$, the classical identity

$$[(u-1)!u!][(v-1)!v!][(m-1)!m!] = [m(u-1)!(v-1)!(m-1)!]^2 \quad (6.2)$$

supplies at most six factorials. Equal indices can be cancelled in pairs; they remove square factors, retain the largest index m , and leave distinct positive indices. At least two remain because $m!$ is not a square for $m > 1$, again by Bertrand's postulate. Thus every composite endpoint has $F(m) \leq 6$.

For clarity, the full exception ledger is

$$\begin{aligned} \#((\mathcal{B}_{1/100} \setminus D_6) \cap [1, X]) &\leq \pi(X) + E_H(X) + O(1) \\ &\quad + O\left(X^{199/200}(\log X)^2\right) + O\left(X^{7/8} \log X\right), \end{aligned} \quad (6.3)$$

where $E_H(X) = O(X/\log X)$ is the one endpoint exceptional set from Harman. Hence the left-hand side is $o(X)$ (indeed $O(X/\log X)$ with the cited exceptional bound). Subtracting it from (3.2) proves (1.3) and the positive-lower-density conclusion. The $O(X/\log X)$ estimate applies to the exceptional difference $\mathcal{B}_{1/100} \setminus D_6$; Proposition 3.5 supplies only an $o(X)$ error for the count of $\mathcal{B}_{1/100}$ itself. No $O(X/\log X)$ remainder is asserted in (1.3). $\square$

Remark 6.3 (Why the exponent matters). This parameter scheme requires $\theta < \eta$ and $3\alpha + 8\eta < 1$. It therefore needs an available almost-all prime-interval exponent below $1/8$. Harman's $1/10 + \varepsilon$ range supplies that margin. A theorem available only above $2/15$ would not close the same parameter choice.

7 Order of growth of the other endpoint classes

For $k \geq 2$ put $D_k = \{m > 1 : F(m) = k\}$ and $D_k(X) = \#(D_k \cap [1, X])$. Let $\mathcal{Q}$ be the set of distinct values $q_a \neq 1$ with $a \geq 1$, and put

$$\kappa_3 = \sum_{q \in \mathcal{Q}} q^{-1/2} = 2.70975 \dots$$

The series converges by Lemma 3.1: a value first attained at index a is at least $\exp(a/3)$ for large a .

Theorem 7.1. (i) For every $\varepsilon > 0$,

$$D_3(X) = \kappa_3 \sqrt{X} + O_\varepsilon(X^{2/5+\varepsilon}).$$

In particular $D_3(X) \asymp \sqrt{X}$, and D_3 has density zero.

$$(ii) \liminf_{X \rightarrow \infty} \frac{D_4(X)}{X} \geq \frac{1}{4}.$$

$$(iii) \liminf_{X \rightarrow \infty} \frac{D_5(X)}{X} \geq \frac{\log 2}{40000}.$$

$$(iv) \liminf_{X \rightarrow \infty} \frac{D_6(X)}{X} \geq \delta_A \log(100/99).$$

Consequently $D_k(X) \asymp X$ for $k = 4, 5, 6$.

Part (iv) is Theorem 1.1. The constants in (ii) and (iii) are not optimized. For (i) and (iii), see also [8].

Remark 7.2 (Formal verification). Theorem 7.1 has been verified in Lean 4 with Mathlib, on top of the formalization of Theorem 1.1 and with no further hypotheses. The only axioms used are `propext`, `Classical.choice` and `Quot.sound`. The inputs of Section 2, Lemma 3.1, Lemma 4.1 and Proposition 5.1 enter through the same formal theorems as in the proof of Theorem 1.1. The Lean development is available at https://github.com/jon-hartley/erdos374_lean_verification.

7.1 Endpoints with nonconsecutive top indices

As in (1.4), a three-factor product $a!b!m!$ with $a < b < m$ and $h = m - b$ is a square exactly when $q_a P_h(m)$ is a square; a two-factor product $b!m!$ is the case $a = 0$. Let $\mathcal{E}$ be the set of m such that $q_a P_h(m)$ is a square for some $a \geq 0$ and $h \geq 2$ with $a + h < m$, and put $E(X) = \#(\mathcal{E} \cap [1, X])$.

Lemma 7.3. *Let $e_1, e_2, d \geq 1$. Then*

$$\#\{u \leq X : e_1 u^2 - e_2 v^2 = d \text{ for some } v \in \mathbb{N}\} \leq d^2 (\log_2(2\sqrt{e_1} X) + 1).$$

Proof. For a solution put $\alpha = \sqrt{e_1} u + \sqrt{e_2} v$. Then $\alpha(\sqrt{e_1} u - \sqrt{e_2} v) = d$, so α determines (u, v) , and $1 \leq \alpha \leq 2\sqrt{e_1} X$. Let (u', v') be a second solution with $u \equiv u' \pmod{d}$ and $v \equiv v' \pmod{d}$, and α' its value. Then $x = (e_1 u u' - e_2 v v')/d$ and $y = (v u' - u v')/d$ are integers. The identity $(e_1 u u' - e_2 v v')^2 - e_1 e_2 (v u' - u v')^2 = (e_1 u^2 - e_2 v^2)(e_1 u'^2 - e_2 v'^2)$ gives $x^2 - e_1 e_2 y^2 = 1$, and $\alpha/\alpha' = x + y\sqrt{e_1 e_2}$. If $x + y\sqrt{e_1 e_2} > 1$, its conjugate lies in $(0, 1)$, so $x, y \geq 1$ and $x + y\sqrt{e_1 e_2} \geq 2$. Hence the integers $\lfloor \log_2 \alpha \rfloor$ are distinct within each of the at most d^2 residue classes of (u, v) modulo d . $\square$

Lemma 7.4. $E(X) = O_\varepsilon(X^{2/5+\varepsilon})$ for every $\varepsilon > 0$.

Proof. Fix a small $\eta > 0$, put $H = X^\eta$, and let $m \leq X$ with a, h as above. Since $q_a \mid P_h(m)$, Lemma 3.1 gives $a \leq 3h \log X + O(1)$. If $h \geq X^{\eta/2}$, Lemma 4.1 with $\gamma = \eta/2$ supplies a prime $p > a$ with $v_p(P_h(m)) = 1$. That prime does not divide q_a , a contradiction. Hence $h < X^{\eta/2}$ and $a < X^\eta$, uniformly and with no exceptional endpoints. Put $\sigma = 1/5$.

Small kernel, $q_a \leq X^{\sigma h}$. A prime $p > h$ divides at most one of $m, \dots, m - h + 1$. If it divides one of them to an odd power, it divides q_a . Bounding the primes $p \leq h$ directly gives

$$\prod_{i < h} \text{sf}(m - i) \leq q_a h^h 4^h.$$

Hence two block elements $m - i = e_1 u^2$ and $m - j = e_2 v^2$ ($i < j < h$) have $e_1 e_2 \leq 16h^2 X^{2\sigma}$ and $e_1 u^2 - e_2 v^2 = j - i \in [1, H]$. By Lemma 7.3 each such equation has $O(H^2 \log X)$ solutions with $u \leq X$. Summing over i, j and over $e_1 e_2 \leq 16H^2 X^{2\sigma}$ bounds the number of these endpoints by $X^{2\sigma+O(\eta)}$.

Large kernel, $q_a > X^{\sigma h}$. Fix (a, h) and let $\mathcal{T}$ be the set of $m \leq X$ with $q_a P_h(m)$ a square. For each prime $p \mid q_a$ with $p > h$, every $m \in \mathcal{T}$ lies in one of h residue classes modulo p . For each odd prime $p \leq Y$, by (2.4) with the zero residues included, the values m for which $q_a P_h(m)$ is a square or zero modulo p occupy at most $(p + h + 5h\sqrt{p})/2$ classes. Gallagher's larger sieve [9] then bounds $\#\mathcal{T}$, and Mertens' theorem with $Y = X^{(1-\sigma)/2+O(\eta)}$ gives $\#\mathcal{T} \leq X^{(1-\sigma)/2+O(\eta)}$.

Both parts are $X^{2/5+O(\eta)}$ for $\sigma = 1/5$, and there are $O(X^{2\eta})$ pairs (a, h) . Letting $\eta \rightarrow 0$ proves the lemma. $\square$

7.2 Proof of Theorem 7.1

Consecutive top indices. When $b = m - 1$, the product $a!(m - 1)!m!$ is a square exactly when $q_a m$ is a square. Let $\mathcal{C}$ be the set of m for which $q_a m$ is a square for some $a < m$. Each $m \in \mathcal{C}$ has the form $q_a t^2$, so by Lemma 3.1 $\#(\mathcal{C} \cap [1, X]) \leq \sum_{a: q_a \leq X} \sqrt{X/q_a} \ll \sqrt{X}$.

A two-factor representation $b!m!$ of a nonsquare m has $m - b \geq 2$, so $m \in \mathcal{E}$. A three-factor representation puts m in $\mathcal{C} \cup \mathcal{E}$.

(i) By the preceding paragraph, $D_3 \subseteq \mathcal{C} \cup \mathcal{E}$, and squares are not in D_3 . Conversely, let $m = qu^2$ with $q \in \mathcal{Q}$ first attained at index a . Then $a!(m - 1)!m!$ is a square, and $a < m - 1$ for all but finitely many such m . Since m is not a square, $m \in D_3$ unless $m \in \mathcal{E}$. Distinct q give disjoint families, since $q = \text{sf}(m)$. Therefore

$$D_3(X) = \sum_{q \in \mathcal{Q}} \lfloor \sqrt{X/q} \rfloor + O(E(X)) + O(1) = \kappa_3 \sqrt{X} + O(\log X) + O(E(X)).$$

The second step uses Lemma 3.1 twice: there are $O(\log X)$ values $q \leq X$, and $\sum_{q > X} q^{-1/2} = O(X^{-1/2} \log X)$. Lemma 7.4 gives (i).

(ii) For $m = 4u$ with $u \geq 2$,

$$(u-1)! u! (m-1)! m! = (2u(u-1)! (m-1)!)^2,$$

so $F(m) \leq 4$. A nonsquare m with $F(m) \in \{2, 3\}$ lies in $\mathcal{C} \cup \mathcal{E}$, which has density zero. Nonsquare multiples of 4 have density $1/4$.

(iii) Let $\mathcal{T}_5$ be the set of even squarefree m with $P^+(m) > m^{99/100}$. For $m = 2z$ with $z \geq 4$,

$$2! (z-1)! z! (2z-1)! (2z)! = (2z(z-1)! (2z-1)!)^2,$$

so $F(m) \leq 5$. Now argue as in Section 6.1 with $\alpha = 1/100$, $\theta = 11/100$, $\eta = 3/25$. Discard the exceptional endpoints of Lemma 6.1 and finitely many initial values.

- The prime $P = P^+(m)$ excludes two- and three-factor representations.
- By Corollary 6.2, both gaps of a four-factor representation are below $m^{3/25}$.
- The paired case $(h, \ell) = (1, 1)$ would make cm a square with $c < m$, which is impossible for squarefree m (Lemma 3.3, multiplier 1).
- Proposition 5.1, which does not assume squarefreeness, bounds the nonpaired four-factor case by $O(X^{7/8} \log X)$.

Hence $F(m) = 5$ for all but $o(X)$ members of $\mathcal{T}_5 \cap [1, X]$. Write $m = rP$ with $r < m^{1/100}$ even and squarefree. Chebyshev's bounds, the positive density of even squarefree numbers, and partial summation give $\#(\mathcal{T}_5 \cap [1, X]) \geq (\log 2/40000) X$ for large X .

(iv) This is Theorem 1.1. □

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